

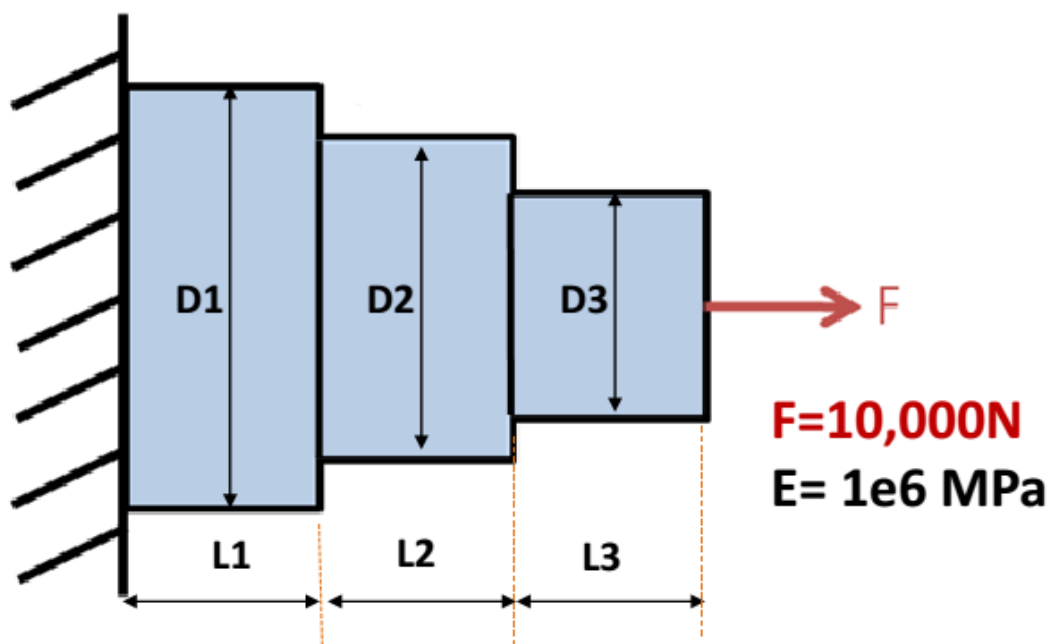
Elastic behavior of simple elements under axial loading.

Using Ansys, it is required:

The 3-step bar is loaded with force F as shown below. Young's modulus E is the same for all steps. The cross-sectional areas are shown in the figure.

Determine using both (APDL and WB):

- The nodal displacements.
- The stresses in each bar element.



$$L1=L2=L3=40$$

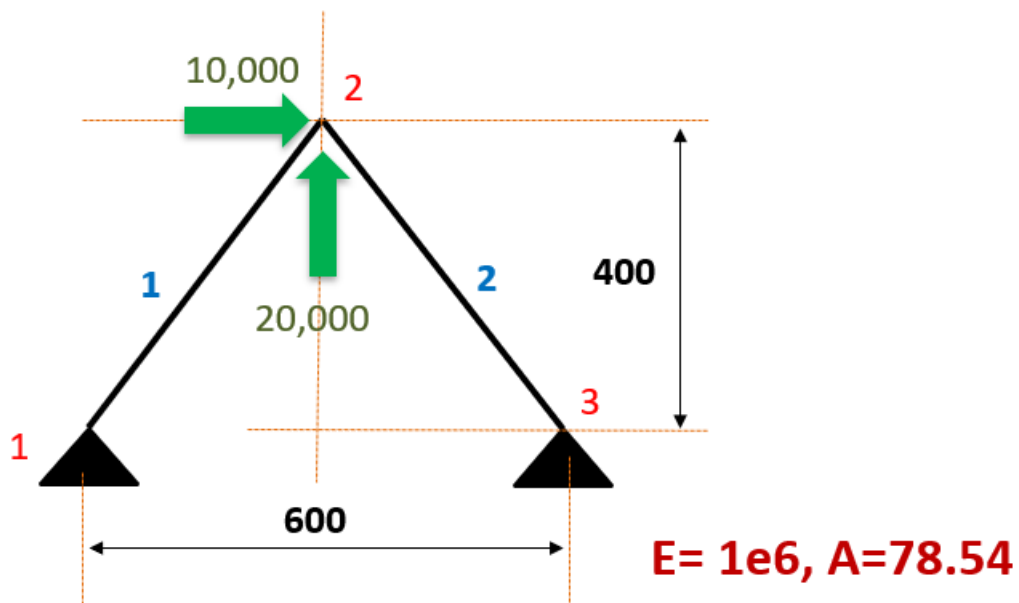
$$D1=50, D2=40, D3=30$$

Using Ansys, it is required:

The plane truss is loaded with F_x and F_y as shown below. Young's modulus E is the same for all the bars. The cross-sectional areas are shown in the figure.

Determine using both (APDL and WB):

- The nodal displacements.
- The stresses in each bar element.

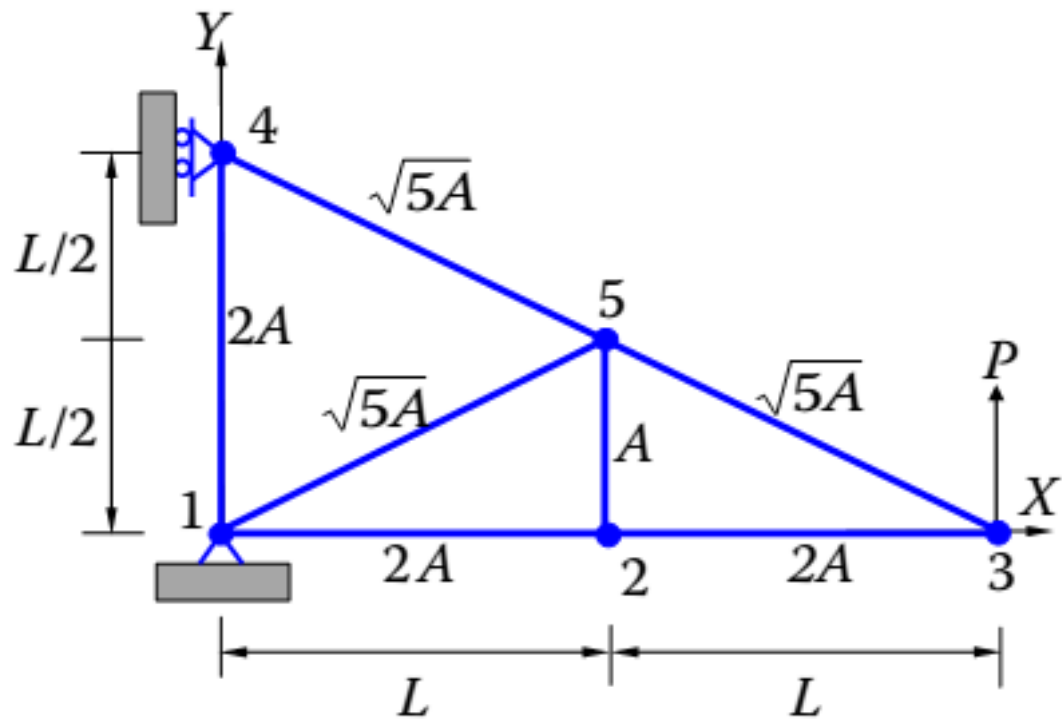


Using Ansys, it is required:

The plane truss is loaded with force P as shown below. Young's modulus E is the same for all the bars. The cross-sectional areas are shown in the figure. Assume (A) as you like.

Determine using both (APDL and WB):

- a. The nodal displacements.
- b. The stresses in each bar element.

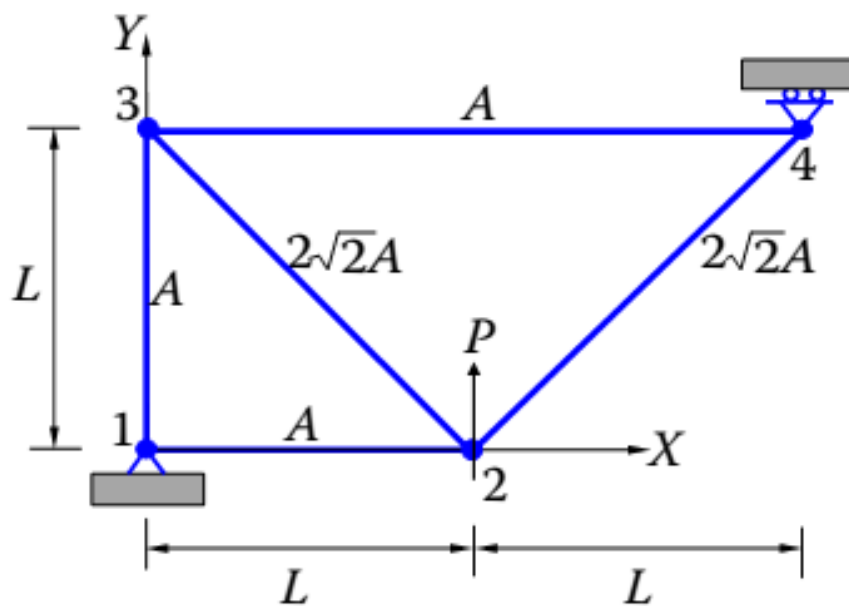


Using Ansys, it is required:

The plane truss is loaded with force P as shown below. Young's modulus E is the same for all the bars. The cross-sectional areas are shown in the figure. Assume (A) as you like.

Determine using both (APDL and WB):

- a. The nodal displacements.
- b. The stresses in each bar element.



Using Ansys, it is required:

The plane truss is loaded with forces 10KN and 20KN as shown below. Young's modulus E is the same for all the bars. The cross-sectional areas are shown in the figure.

Determine using both (APDL and WB):

- The nodal displacements.
- The stresses in each bar element.

